



Assignment No.9

OBJECT ORIENTED PROGRAMMING

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2034

1. Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

CODE:

```
#include <iostream>

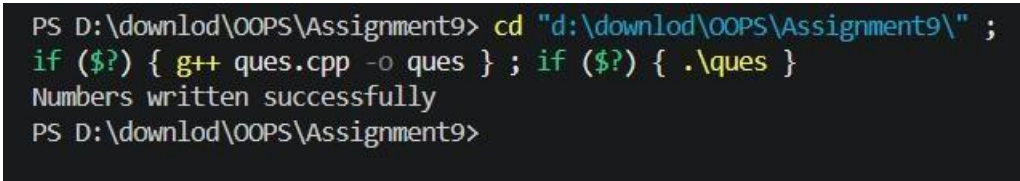
#include <fstream> using
namespace std;

int main() {    ofstream
file("NUM.TXT");

    for(int i = 1; i <= 200; i++) {
file << i << " ";
    }

    file.close();

    cout << "Numbers written successfully";
}
```



```
PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\" ;
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }
Numbers written successfully
PS D:\download\OOPS\Assignment9>
```

2. Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

CODE:

```
#include <iostream>

#include <fstream>

#include <cctype> using
namespace std; void
```

```

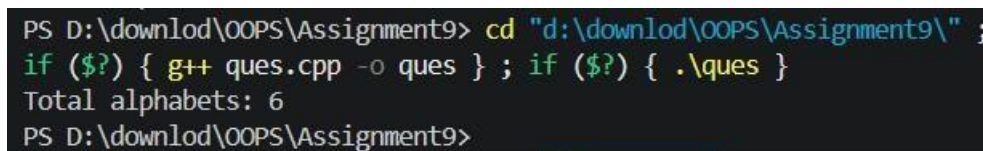
countAlphabets() {
ifstream
file("NOTES.TXT");
    if(!file) {
        cout << "File not found";
        return;
    }
    char ch;
int count = 0;

    while(file.get(ch)) {
if(isalpha(ch))
count++;
    }
    cout << "Total alphabets: " << count;
    file.close();
}
int main() {
countAlphabets();
}

```



A screenshot of a text editor window. The title bar says 'NOTES.TXT'. The first line of the file contains the word 'Helloo'.



A screenshot of a Windows command prompt. The user has navigated to 'D:\download\OOPS\Assignment9' and compiled a C++ program named 'ques.cpp' into an executable named 'ques'. They then ran the program, which output 'Total alphabets: 6'.

```

PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\" ;
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }
Total alphabets: 6
PS D:\download\OOPS\Assignment9>

```

3. Write a program to copy the contents of one file to another.

CODE:

```

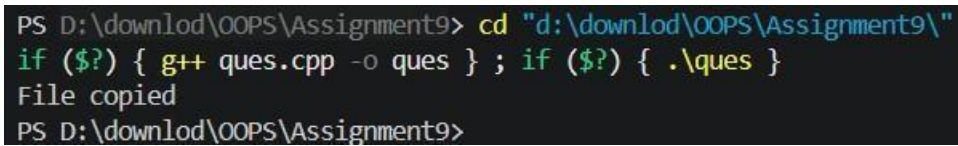
#include <iostream>

#include <fstream>

```

```
using namespace std;
```

```
int main() { ifstream  
fin("input.txt"); ofstream  
fout("output.txt");  
char ch;  
  
while(fin.get(ch)) {  
fout << ch;  
}  
  
cout << "File copied";  
fin.close();  
fout.close();  
}
```



```
PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\"  
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }  
File copied  
PS D:\download\OOPS\Assignment9>
```

4. Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

CODE:

```
#include <iostream>  
#include <fstream>  
#include <cstring> using  
namespace std;
```

```
int main() {  
char str[100];  
cout << "Enter  
string: ";
```

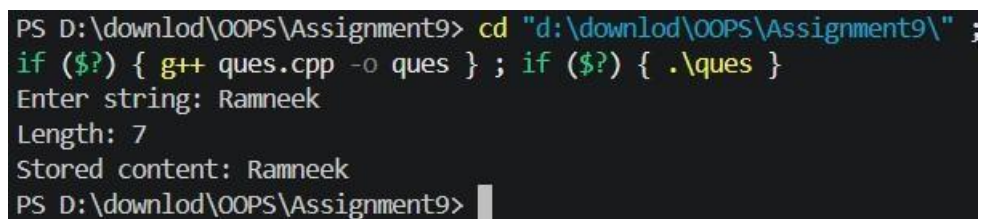
```

cin.getline(str,
100);    cout <<
"Length: " <<
strlen(str)    <<
endl;

ofstream
file("data.txt");
file    <<    str;
file.close();

ifstream
file2("data.txt")
;        char ch
cout << "Stored
content:    ";
while(file2.get(
ch)) {        cout
<< ch;
    }
    file2.close();
}

```



```

PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\" ;
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }
Enter string: Ramneek
Length: 7
Stored content: Ramneek
PS D:\download\OOPS\Assignment9>

```

5. Write a program to do the followings:

- Create a file A-Z and read 10 th character using seekg().
- Overwrite 5 th number using seekp().
- Find file size using tellg().
- Read last character using seekg().

CODE:

```

#include <iostream>

#include <fstream> using
namespace std;

int main() { fstream file("file.txt", ios::out | ios::in |
ios::trunc); // (a) Write A-Z for(char ch = 'A'; ch <=
'Z'; ch++) {
    file << ch;
}
file.flush();
// Read 10th character
file.seekg(9, ios::beg);
char ch; file.get(ch);
cout << "10th char: " << ch << endl;

// (b) Overwrite 5th character
file.seekp(4, ios::beg); file <<
'#';

// (c) File size file.seekg(0,
ios::end); cout << "Size: " <<
file.tellg() << endl;

// (d) Last character file.seekg(-
1, ios::end); file.get(ch); cout <<
"Last char: " << ch << endl;

file.close();
}

```

```

PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\"
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }
10th char: J
Size: 26
Last char: Z
PS D:\download\OOPS\Assignment9> 

```

(e) Create a file data.txt with multiple lines. Read the file and:

i. Move the pointer to the 10 th byte using seekg()

ii. Display the current position using tellg(). iii.

Read and display the remaining content.

CODE:

```
#include <iostream>
```

```
#include <fstream> using
```

```
namespace std;
```

```
int main() { ofstream out("data.txt"); out << "This is a sample
```

```
file\nWith multiple lines\nFor testing";
```

```
out.close();
```

```
ifstream in("data.txt");
```

```
in.seekg(10); cout << "Position: " <<
```

```
in.tellg() << endl;
```

```
char ch;
```

```
while(in.get(ch)) {
```

```
cout << ch;
```

```
}
```

```
in.close();
```

```
}
```

```

PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\"
if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }
Position: 10
sample file
with multiple lines
For testing
PS D:\download\OOPS\Assignment9>

```

6. Write a program to do the followings:

- i. Create a file with text "Hello World"
- ii. After writing each character, display the current position of the put pointer using tellp().
- iii. Replace "World" with "C++" using random access

CODE:

```

#include <iostream>

#include <fstream> using
namespace std;

int main() {    fstream file("hello.txt", ios::out | ios::in
| ios::trunc);

    if(!file) {    cout <<
"File error";    return
0;
    }

    string str = "Hello World";    //
write + show position    for(int i =
0; i < str.length(); i++) {

        file << str[i];

        cout << "Position: " << file.tellp() << endl;

    }
}

```



```
// replace "World" with "C++"  
  
file.seekp(6);  
file << "C++";
```

```
file.close();
```

```
// show final content  
ifstream in("hello.txt");  
  
char ch;  
while(in.get(ch)) {  
    cout << ch;  
}  
}
```

```
PS D:\download\OOPS\Assignment9> cd "d:\download\OOPS\Assignment9\" ; if ($?) { g++ ques.cpp -o ques } ; if ($?) { .\ques }  
Position: 1  
Position: 2  
Position: 3  
Position: 4  
Position: 5  
Position: 6  
Position: 7  
Position: 8  
Position: 9  
Position: 10  
Position: 11  
Hello C++ld  
PS D:\download\OOPS\Assignment9> 
```